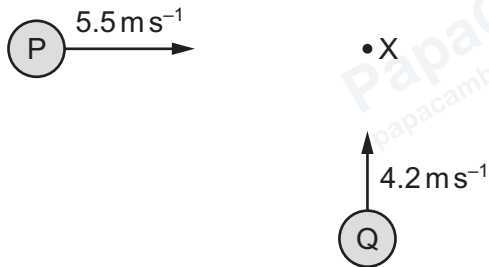


12 The diagram shows the view from above of two hard balls rolling along a horizontal table.



The balls each have a mass of 0.16 kg .

The initial velocity of ball P is 5.5 m s^{-1} and the initial velocity of ball Q is 4.2 m s^{-1} .

The initial velocity of ball P is perpendicular to the initial velocity of ball Q.

The balls collide at point X where ball Q stops immediately.

There are no resistive forces acting on the balls.

What is the speed of ball P after the collision?

- A** 1.3 m s^{-1} **B** 5.5 m s^{-1} **C** 6.9 m s^{-1} **D** 9.7 m s^{-1}